

Rodrigo Hiroto Morais

+55 (11) 95841-7956 | rodrigohmorais@proton.me | linkedin.com/in/rodrigohmorais | github.com/rodigu | rodigu.github.io

Data Engineer with 5 years of experience building and maintaining large-scale data pipelines, ETL systems, and cloud data infrastructure. Proven track record of owning data architecture end-to-end from ingestion and transformation to quality assurance and BI enablement. Experienced in deploying LLM-powered pipelines and medallion architectures on Azure, with a research background in NLP and network science.

Experience

Data Engineer | CEJAM - Centro de Estudos e Pesquisas Dr. João Amorim | São Paulo, Brazil | *January 2026 – Present*

- Deploy LLMs to Azure AI Foundry, integrating them into Databricks pipelines for NLP information extraction on unstructured data
- Design and maintain medallion architecture (bronze/silver/gold) for ingestion and transformation from heterogeneous sources into analytics-ready databases
- Build ETL pipelines with Databricks, Azure Data Factory, and Azure Data Lake Gen 2
- Manage and optimize Azure cloud infrastructure costs and resources

Data Scientist / Data Engineer | Star Parks | São Paulo, Brazil | *October 2024 – January 2026*

- Designed and maintained business-critical ETL pipelines on Airflow, ensuring reliability and scalability of data workflows
- Implemented modular SQL data models using SQLAlchemy and PyODBC for HTTP API integrations
- Managed Airflow DAG configuration and versioning through Git and GitHub
- Led Airflow 2 to 3 migration and deployment on Azure
- Built PowerBI dashboards enabling data-driven business decisions across teams

Applied Data Science Researcher | University of Illinois Urbana-Champaign | Champaign, Illinois | *August 2022 – August 2024*

- Researched data analysis methods on multi-modal educational datasets
- Developed NLP pipelines using LLMs for qualitative coding of large-scale textual data
- Integrated the Ollama API with Python to operationalize LLM-based workflows

Mathematics Research Assistant | Soka University of America | Aliso Viejo, California | *September 2021 – May 2022*

- Built TypeScript software to mine and transform data from currency markets APIs
- Designed novel algorithms for detecting unbalanced trade cycles in currency exchange networks

Software Developer | PushStart Studio | São Paulo, Brazil | *March 2021 – June 2021*

- Developed web-based applications using JavaScript and PixiJS
- Coordinated with a cross-functional team via Git, Slack, and JIRA

Selected Projects

- **Graph Simulation of Food Banks** | *Python, Pandas, NumPy, NetworkX*
 - Monte Carlo simulation on real food bank data from Illinois to model the impact of catastrophic disruptions on supply chain and distribution networks
- **Machine-Learning for Supernovae Age Detection** | *Python, Pandas, SciKit-Learn, Jupyter*
 - Novel ML pipeline in partnership with the UIUC Astronomy department to classify supernovae age from light band readings
- **PyDBCon** | *Python, Pandas, SQL* — Open source library to streamline API-to-SQL database integrations
- **ZenDBCon** | *Python, Pandas, SQL* — Open source library to integrate the Zendesk API into SQL databases

Certifications & Honors

DS4A 2.0 Honors | Sept 2021

- Merit fellowship for highly selective data science program (4% acceptance rate) under professor Natesh Pillai (Harvard)
- Delivered insights on user behavior from a dataset of 200K+ users for an EdTech company

SoftBank Endowed Scholar | Sept 2021

Skills

Languages & Tools: Python, SQL, Airflow, Databricks, Apache Spark, Azure Data Factory, Azure Data Lake Gen 2, Azure AI Foundry, PowerBI, DAX, Polars, Pandas, NumPy, SQLAlchemy, FastAPI, Git, Linux, TypeScript, JavaScript, R

Data Engineering: ETL Pipeline Design, Medallion Architecture, Data Lake & Data Warehouse, Data Quality & Monitoring, Pipeline Scalability, BI Enablement, LLM Integration, NLP, REST API Integrations, Cloud Infrastructure (Azure)

Education

University of Illinois Urbana-Champaign | *August 2022 – May 2024* PhD in Informatics (withdrawn)

Soka University of America | Aliso Viejo, California | *August 2018 – May 2022* Bachelor's degree, Liberal Arts and Sciences — concentrations in Economics and International Studies